

Money in the Utility Function

Government

$$\begin{aligned} T_t &= M_t - M_{t-1} \\ &= \mu_t M_{t-1} - M_{t-1} \\ &= (\mu_t - 1) M_{t-1} \end{aligned}$$

$$\ln \mu_{t+1} = (1 - \alpha) \ln \mu + \alpha \ln \mu_t + \eta_{t+1}$$

where $t \in \{0, 1, \dots\}$, T_t are lump-sum transfers during period t , M_t is the money supply at the end of t , the initial stock $M_{-1} > 0$ is given, μ_t is the gross growth rate of the money supply, $\alpha \in (0, 1)$, $\mu > 0$, and the η_t 's are *iid* with zero mean and variance σ_η^2 .

Technology

$$Y_t = Z_t K_t^\theta H_t^{1-\theta}, \quad \theta \in (0, 1)$$

$$\ln Z_{t+1} = (1 - \gamma) \ln Z + \gamma \ln Z_t + \varepsilon_{t+1}$$

with $\gamma \in (0, 1)$, $Z > 0$, and the ε_t 's are *iid* with zero mean and variance σ_ε^2 .

We assume that the η_t 's and ε_t 's are independent from each other at all leads and lags.

Remark on Notation

Capital letters are used to distinguish per capita variables that a competitive household takes as parametric from individual-specific variables that are chosen by the household (in equilibrium these will be the same). Aggregate variables and per capita variables coincide because there's a mass-one continuum of (identical) households.

Representative Household

Expected Utility

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u \left(c_t, h_t, \frac{m_t}{p_t} \right)$$

where

$$u \left(c_t, h_t, \frac{m_t}{p_t} \right) = \ln c_t - B h_t + a \ln \left(\frac{m_t}{p_t} \right)$$

with $\beta \in (0, 1)$, $B > 0$, $a > 0$. We assume $\beta < \mu$.

Budget Constraint

$$c_t + k_{t+1} + \frac{b_t}{p_t} + \frac{m_t}{p_t} = w_t h_t + r_t k_t + (1 - \delta) k_t \\ + \frac{(1 + i_{t-1}) b_{t-1}}{p_t} + \frac{m_{t-1}}{p_t} + \frac{T_t}{p_t}$$

where we have imposed zero profits (because technology displays CRS), and $m_{-1} = M_{-1} > 0$, $b_{-1} = 0$ and $k_0 > 0$ are given.

Profit Maximization

$$\Pi_t \equiv \max_{K_t, H_t} Z_t K_t^\theta H_t^{1-\theta} - w_t H_t - r_t K_t$$

FOC give:

$$w_t = (1 - \theta) Z_t \left(\frac{K_t}{H_t} \right)^\theta = (1 - \theta) \frac{Y_t}{H_t}$$

$$r_t = \theta Z_t \left(\frac{K_t}{H_t} \right)^{-(1-\theta)} = \theta \frac{Y_t}{K_t}$$

$$\Pi_t = Y_t - w_t H_t - r_t K_t = 0 \quad (\text{zero profits})$$

Utility Maximization

Lagrangian

$$\mathcal{L} = \mathbb{E}_0 \left\{ \sum_{t=0}^{\infty} \beta^t u(c_t, h_t, m_t/p_t) + \sum_{t=0}^{\infty} \beta^t \lambda_t [w_t h_t + r_t k_t + (1-\delta)k_t + \frac{(1+i_{t-1})b_{t-1}}{p_t} + \frac{m_{t-1}}{p_t} + \frac{T_t}{p_t} - c_t - k_{t+1} - \frac{b_t}{p_t} - \frac{m_t}{p_t}] \right\}$$

First-Order Conditions (FOC)

$$c_t : \quad u_c(t) = \lambda_t$$

$$h_t : \quad -u_h(t) = \lambda_t w_t$$

$$m_t : \quad u_m(t) \frac{1}{p_t} - \lambda_t \frac{1}{p_t} + \beta \mathbb{E}_t \left\{ \lambda_{t+1} \frac{1}{p_{t+1}} \right\} = 0$$

$$k_{t+1} : \quad \lambda_t = \beta \mathbb{E}_t \{ \lambda_{t+1} (r_{t+1} + 1 - \delta) \}$$

$$b_t : \quad \lambda_t \frac{1}{p_t} = \beta \mathbb{E}_t \left\{ \lambda_{t+1} \frac{1+i_t}{p_{t+1}} \right\}$$

plus the budget constraint and appropriate transversality conditions. Notice that we use u_m to denote the partial derivative of u with respect to its third argument, $\frac{m}{p}$.

From the FOCs w.r.t. c_t and h_t we get:

$$\frac{-u_h(t)}{u_c(t)} = w_t$$

From the FOCs w.r.t. c_t and k_{t+1} we get:

$$u_c(t) = \beta \mathbb{E}_t \{u_c(t+1)(r_{t+1} + 1 - \delta)\}$$

From the FOCs w.r.t. c_t and b_t we get:

$$\begin{aligned} u_c(t) &= \beta \mathbb{E}_t \{u_c(t+1) (1 + i_t) \frac{p_t}{p_{t+1}}\} \\ &= \beta (1 + i_t) \mathbb{E}_t \{u_c(t+1) \frac{p_t}{p_{t+1}}\} \end{aligned}$$

From the FOCs w.r.t. c_t and m_t we get:

$$u_c(t) = u_m(t) + \beta \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\}$$

Combining the previous two expressions we get:

$$\begin{aligned} u_c(t) &= u_m(t) + \beta \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\} \\ &= u_m(t) + \frac{u_c(t)}{1 + i_t} \end{aligned}$$

Then:

$$\frac{u_m(t)}{u_c(t)} = \frac{i_t}{1 + i_t}$$

Exercise: Interpret.

For $u(c_t, h_t, m_t/p_t) = \ln c_t - Bh_t + a \ln(m_t/p_t)$ we have:

$$u_c = \frac{1}{c}$$

$$u_h = -B$$

$$u_m = \frac{a}{\frac{m}{p}}$$

Define:

$$m_t^r \equiv \frac{m_t}{p_t} \quad (\text{real balances})$$

$$\pi_{t+1} \equiv \frac{p_{t+1}}{p_t} \quad (\text{gross inflation rate})$$

Then, $\frac{-u_h(t)}{u_c(t)} = w_t$ becomes:

$$Bc_t = w_t$$

From $u_c(t) = \beta \mathbb{E}_t \{u_c(t+1)(r_{t+1} + 1 - \delta)\}$ we get:

$$\frac{1}{\beta} = \mathbb{E}_t \left\{ \frac{c_t}{c_{t+1}} (r_{t+1} + 1 - \delta) \right\}$$

From $u_c(t) = u_m(t) + \beta \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\}$ we get:

$$\frac{1}{c_t} = \frac{a}{m_t^r} + \beta \mathbb{E}_t \left\{ \frac{1}{c_{t+1}} \frac{1}{\pi_{t+1}} \right\}$$

From $\frac{u_m(t)}{u_c(t)} = \frac{i_t}{1+i_t}$ we get:

$$m_t^r = a \frac{1 + i_t}{i_t} c_t$$

Equilibrium requires:

$$c_t = C_t$$

$$k_t = K_t$$

$$h_t = H_t$$

$$m_t = M_t$$

$$b_t = B_t = 0$$

Using the conditions above and the government's budget constraint into the budget constraint of the household we get:

$$C_t + K_{t+1} + \frac{B_t}{p_t} + \frac{M_t}{p_t} = w_t H_t + r_t K_t + (1 - \delta) K_t + \frac{(1 + i_{t-1}) B_{t-1}}{p_t} + \frac{M_{t-1}}{p_t} + \frac{(\mu_t - 1) M_{t-1}}{p_t}$$

Since $I_t = K_{t+1} - (1 - \delta) K_t$ (gross investment), $\mu_t M_{t-1} = M_t$, and $w_t H_t + r_t K_t = Y_t$ (zero profits), we get:

$$C_t + I_t = Y_t$$

In addition, from $M_t = \mu_t M_{t-1}$ we get: $\frac{M_t}{p_t} = \mu_t \frac{M_{t-1}}{p_t}$.

Then: $\frac{M_t}{p_t} = \mu_t \frac{M_{t-1}}{p_{t-1}} \frac{p_{t-1}}{p_t} \Rightarrow$

$$M_t^r = \frac{\mu_t}{\pi_t} M_{t-1}^r$$

Summarizing (12 eqs., 12 var.):

$$Y_t = Z_t K_t^\theta H_t^{1-\theta}$$

$$w_t = (1 - \theta) \frac{Y_t}{H_t}$$

$$r_t = \theta \frac{Y_t}{K_t}$$

$$I_t = K_{t+1} - (1 - \delta)K_t$$

$$Y_t = C_t + I_t$$

$$M_t^r = a^{\frac{1+i_t}{i_t}} C_t$$

$$BC_t = w_t$$

$$M_t^r = \frac{\mu_t}{\pi_t} M_{t-1}^r$$

$$\frac{1}{\beta} = \mathbb{E}_t \left\{ \frac{C_t}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\}$$

$$\frac{1}{C_t} = \frac{a}{M_t^r} + \beta \mathbb{E}_t \left\{ \frac{1}{C_{t+1}} \frac{1}{\pi_{t+1}} \right\}$$

$$\ln Z_{t+1} = (1 - \gamma) \ln Z + \gamma \ln Z_t + \varepsilon_{t+1}$$

$$\ln \mu_{t+1} = (1 - \alpha) \ln \mu + \alpha \ln \mu_t + \eta_{t+1}$$

Nonstochastic Steady State

We have:

$$\bar{\mu} = \mu$$

$$\bar{Z} = Z$$

From $M_t^r = \frac{\mu_t}{\pi_t} M_{t-1}^r$ we get: $\bar{M}^r = \frac{\bar{\mu}}{\bar{\pi}} \bar{M}^r$. Then:

$$\begin{aligned}\bar{\pi} &= \bar{\mu} \\ &= \mu\end{aligned}$$

From $\frac{1}{\beta} = \mathbb{E}_t \left\{ \frac{C_t}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\}$, $C_t = C_{t+1}$, and $r_{t+1} = \bar{r}$ we get:

$$\bar{r} = \frac{1}{\beta} - 1 + \delta$$

From $r_t = \theta \frac{Y_t}{K_t} = \theta Z_t \left(\frac{K_t}{H_t} \right)^{-(1-\theta)}$ we get: $\bar{r} = \theta \bar{Z} \left(\frac{\bar{K}}{\bar{H}} \right)^{-(1-\theta)}$.

Then:

$$\frac{\bar{K}}{\bar{H}} = \left(\frac{\theta \bar{Z}}{\bar{r}} \right)^{\frac{1}{1-\theta}}$$

From $w_t = (1 - \theta) \frac{Y_t}{H_t} = (1 - \theta) Z_t \left(\frac{K_t}{H_t} \right)^\theta$ we get:

$$\bar{w} = (1 - \theta) \bar{Z} \left(\frac{\bar{K}}{\bar{H}} \right)^\theta$$

From $BC_t = w_t$ we get:

$$\bar{C} = \frac{\bar{w}}{B}$$

From $\frac{1}{C_t} = \frac{a}{M_t^r} + \beta \mathbb{E}_t \left\{ \frac{1}{C_{t+1}} \frac{1}{\pi_{t+1}} \right\}$ we get: $\frac{1}{C} = \frac{a}{\bar{M}^r} + \beta \frac{1}{C} \frac{1}{\bar{\pi}}$. Then, using $\bar{\pi} = \mu$, we get:

$$\bar{M}^r = a \bar{C} \frac{\mu}{\mu - \beta}$$

Recall we have assumed $\mu > \beta$. Then, $\bar{M}^r > 0$.

From $M_t^r = a \frac{1+i_t}{i_t} C_t$ we get: $\overline{M}^r = a \frac{1+\bar{i}}{\bar{i}} \overline{C}$. Using the previous expression to eliminate $\overline{M}^r$ we get: $a \overline{C} \frac{\mu}{\mu-\beta} = a \frac{1+\bar{i}}{\bar{i}} \overline{C}$. Then: $\frac{\mu}{\mu-\beta} = \frac{1+\bar{i}}{\bar{i}}$. Then:

$$\bar{i} = \frac{\mu - \beta}{\beta}$$

Since we have assumed $\mu > \beta$, we get $\bar{i} > 0$.

From the expression above we get:

$$\begin{aligned} 1 + \bar{i} &= \frac{1}{\beta} \mu \\ &= \frac{1}{\beta} \pi \end{aligned}$$

This shows that the gross real interest rate in the nss is $\frac{1}{\beta}$, which is consistent with the expression we found earlier for the rental rate of capital: $\bar{r} - \delta = \frac{1}{\beta} - 1$.

Now we express the rest of the variables in terms of $\bar{K}$ and then solve for $\bar{K}$.

From $I_t = K_{t+1} - (1 - \delta)K_t$ we get:

$$\bar{I} = \delta \bar{K}$$

From $\frac{\bar{K}}{\bar{H}} = \left(\frac{\theta \bar{Z}}{\bar{r}}\right)^{\frac{1}{1-\theta}}$ we get:

$$\bar{H} = \left(\frac{\theta \bar{Z}}{\bar{r}}\right)^{\frac{-1}{1-\theta}} \bar{K}$$

From $Y_t = Z_t K_t^\theta H_t^{1-\theta}$ we get: $\bar{Y} = \bar{Z} \bar{K}^\theta \bar{H}^{1-\theta}$. Using the expression above: $\bar{Y} = \bar{Z} \bar{K}^\theta \left(\frac{\theta \bar{Z}}{\bar{r}}\right)^{-1} \bar{K}^{1-\theta}$. Then:

$$\bar{Y} = \frac{\bar{r}}{\theta} \bar{K}$$

From $Y_t = C_t + I_t$ we get: $\bar{Y} = \bar{C} + \bar{I}$. Using the previous expressions: $\frac{\bar{r}}{\theta} \bar{K} = \frac{\bar{w}}{B} + \delta \bar{K}$. Then:

$$\bar{K} = \frac{\bar{w}}{\left(\frac{\bar{r}}{\theta} - \delta\right) B}$$

where $\bar{K} > 0$ because $\bar{r} > \delta$ and $\theta \in (0, 1)$.

Notice that the only real variable that depends on μ in the nss is $\bar{M}^r$. This is because the nominal interest rate $\bar{i}$ increases with $\bar{\pi} = \mu$.

Log-linearization

For any variable X define:

$$\widetilde{X}_t \equiv \ln X_t - \ln \bar{X}$$

Using the standard method to log-linearize our system, we obtain:

$$\widetilde{Y}_t = \widetilde{Z}_t + \theta \widetilde{K}_t + (1 - \theta) \widetilde{H}_t$$

$$\widetilde{w}_t = \widetilde{Y}_t - \widetilde{H}_t$$

$$\widetilde{r}_t = \widetilde{Y}_t - \widetilde{K}_t$$

$$\delta \widetilde{I}_t = \widetilde{K}_{t+1} - (1 - \delta) \widetilde{K}_t$$

$$\bar{Y} \widetilde{Y}_t = \bar{C} \widetilde{C}_t + \bar{I} \widetilde{I}_t$$

$$\widetilde{M}_t^r = \widetilde{C}_t - \frac{\beta}{\mu} \widetilde{i}_t$$

$$\tilde{C}_t = \tilde{w}_t$$

$$\tilde{M}_t^r = \tilde{M}_{t-1}^r + \tilde{\mu}_t - \tilde{\pi}_t$$

$$0 = \mathbb{E}_t \left\{ \tilde{C}_{t+1} - \tilde{C}_t - \beta \bar{r} \tilde{r}_{t+1} \right\}$$

$$0 = \mathbb{E}_t \left\{ \beta \tilde{C}_{t+1} + \beta \tilde{\pi}_{t+1} - \mu \tilde{C}_t + (\mu - \beta) \tilde{M}_t^r \right\}$$

$$\tilde{Z}_{t+1} = \gamma \tilde{Z}_t + \varepsilon_{t+1} \quad \text{with} \quad \varepsilon_{t+1} \sim \mathcal{N} \left(0, \sigma_\varepsilon^2 \right)$$

$$\tilde{\mu}_{t+1} = \alpha \tilde{\mu}_t + \eta_{t+1} \quad \text{with} \quad \eta_{t+1} \sim \mathcal{N} \left(0, \sigma_\eta^2 \right)$$

Remark. To obtain the second expectational equation use the fact that $\frac{a\bar{C}}{\bar{M}^r} = \frac{\mu-\beta}{\mu}$.

Price Level

From $\pi_{t+1} \equiv \frac{p_{t+1}}{p_t}$ we get: $p_{t+1} = \pi_{t+1}p_t \Rightarrow$

$$\ln p_{t+1} = \ln \pi_{t+1} + \ln p_t.$$

From $\tilde{\pi}_{t+1} = \ln \pi_{t+1} - \ln \bar{\pi}$ and $\bar{\pi} = \mu$ we get:

$$\ln \pi_{t+1} = \ln \mu + \tilde{\pi}_{t+1}.$$

Then:

$$\ln p_{t+1} = \ln \mu + \tilde{\pi}_{t+1} + \ln p_t$$

At $t = 0$, p_0 is determined from the money-market equilibrium condition: $M_0^r = a \frac{1+i_0}{i_0} C_0 \Rightarrow \frac{M_0}{p_0} = a \frac{1+i_0}{i_0} C_0 \Rightarrow$

$$p_0 = \frac{1}{aC_0} \frac{i_0}{1+i_0} M_0$$

Notice that, from the solution of the model, we know $\tilde{C}_0$ and $\tilde{i}_0$. Therefore, we know $C_0 = \bar{C}e^{\tilde{C}_0}$ and $i_0 = \bar{i}e^{\tilde{i}_0}$. Hence, given M_0 , the expression above determines p_0 .

Matrix Form

Define:

$$x_t \equiv \begin{bmatrix} \widetilde{K}_{t+1} \\ \widetilde{M}_t^r \end{bmatrix}, \quad y_t \equiv \begin{bmatrix} \widetilde{Y}_t \\ \widetilde{C}_t \\ \widetilde{I}_t \\ \widetilde{H}_t \\ \widetilde{r}_t \\ \widetilde{w}_t \\ \widetilde{i}_t \\ \widetilde{\pi}_t \end{bmatrix}$$

$$z_t \equiv \begin{bmatrix} \widetilde{Z}_t \\ \widetilde{\mu}_t \end{bmatrix}, \quad \epsilon_t \equiv \begin{bmatrix} \varepsilon_t \\ \eta_t \end{bmatrix}$$

Then, we can rewrite our log-linearized system in matrix form, as follows:

$$\mathbf{0} = Ax_t + Bx_{t-1} + Cy_t + Dz_t$$

$$\mathbf{0} = \mathbb{E}_t\{Fx_{t+1} + Gx_t + Hx_{t-1} + Jy_{t+1} + Ky_t \\ + Lz_{t+1} + Mz_t\}$$

$$z_{t+1} = Nz_t + \epsilon_{t+1} \quad \text{with} \quad \mathbb{E}_t\{\epsilon_{t+1}\} = \mathbf{0}$$

where

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} -\theta & 0 \\ 0 & 0 \\ -1 & 0 \\ -(1-\delta) & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & -(1-\theta) & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -\delta & 0 & 0 & 0 & 0 & 0 \\ \overline{Y} & -\overline{C} & -\overline{I} & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & \frac{\beta}{\mu} & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} -1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -1 \end{bmatrix}$$

$$F = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad G = \begin{bmatrix} 0 & 0 \\ 0 & \mu - \beta \end{bmatrix}, \quad H = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$J = \begin{bmatrix} 0 & 1 & 0 & 0 & -\beta\bar{r} & 0 & 0 & 0 \\ 0 & \beta & 0 & 0 & 0 & 0 & 0 & \beta \end{bmatrix}$$

$$K = \begin{bmatrix} 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\mu & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$L = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad M = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$N = \begin{bmatrix} \gamma & 0 \\ 0 & \alpha \end{bmatrix}$$

Calibration

For quarterly data (similar the the CIA model of Cooley & Hansen 1989):

$$\overline{Z} = 1$$

$$\theta = 0.36$$

$$\beta = 0.99$$

$$B = 2.86$$

$$\delta = 0.025$$

$$\gamma = 0.95$$

$$\alpha = 0.48$$

$$\sigma_{\varepsilon} = 0.00721$$

$$\sigma_{\eta} = 0.009$$

$$a = 0.01$$

Remark. The value of a is chosen so that, when $\mu = 1$, the value of $\overline{M}^r$ is similar to the one implied by the CIA model of Cooley & Hansen (1989). This is the value chosen by McCandless (2008), Chapter 9.

We solve the model for different values of μ :

$$\mu \in \{1.015, 1.15\}$$

This calibration implies:

	$\mu = 1.015$	$\mu = 1.15$
$\bar{r}$	0.0351	0.0351
$\bar{w}$	2.3706	2.3706
$\bar{C}$	0.8288	0.8288
$\bar{K}$	11.4324	11.4324
$\bar{H}$	0.3009	0.3009
$\bar{Y}$	1.1147	1.1147
$\bar{I}$	0.2858	0.2858
$\bar{M}^r$	0.3365	0.0596
$\bar{i}$	0.0253	0.1616
$\bar{\pi}$	1.015	1.15

Solution

$$x_t = Px_{t-1} + Qz_t$$

$$y_t = Rx_{t-1} + Sz_t$$

For $\mu = 1.015$ we get:

$$P = \begin{bmatrix} 0.9418 & 0 \\ 0.5316 & 0 \end{bmatrix}, \quad Q = \begin{bmatrix} 0.1552 & 0 \\ 0.4703 & -0.8803 \end{bmatrix}$$

$$R = \begin{bmatrix} 0.0550 & 0 \\ 0.5316 & 0 \\ -1.3273 & 0 \\ -0.4766 & 0 \\ -0.9450 & 0 \\ 0.5316 & 0 \\ 0 & 0 \\ -0.5316 & 1 \end{bmatrix}, \quad S = \begin{bmatrix} 1.9417 & 0 \\ 0.4703 & 0 \\ 6.2091 & 0 \\ 1.4715 & 0 \\ 1.9417 & 0 \\ 0.4703 & 0 \\ 0 & 0.9026 \\ -0.4703 & 1.8803 \end{bmatrix}$$

Then:

$$\widetilde{K}_{t+1} = 0.9418\widetilde{K}_t + 0.1552\widetilde{Z}_t$$

$$\widetilde{M}_t^r = 0.5316\widetilde{K}_t + 0.4703\widetilde{Z}_t - 0.8803\widetilde{\mu}_t$$

$$\widetilde{Y}_t = 0.0550\widetilde{K}_t + 1.9417\widetilde{Z}_t$$

$$\widetilde{C}_t = 0.5316\widetilde{K}_t + 0.4703\widetilde{Z}_t$$

$$\widetilde{I}_t = -1.3273\widetilde{K}_t + 6.2091\widetilde{Z}_t$$

$$\widetilde{H}_t = -0.4766\widetilde{K}_t + 1.4715\widetilde{Z}_t$$

$$\widetilde{r}_t = -0.9450\widetilde{K}_t + 1.9417\widetilde{Z}_t$$

$$\widetilde{w}_t = 0.5316\widetilde{K}_t + 0.4703\widetilde{Z}_t$$

$$\widetilde{i}_t = 0.9026\widetilde{\mu}_t$$

$$\widetilde{\pi}_t = -0.5316\widetilde{K}_t + \widetilde{M}_{t-1}^r - 0.4703\widetilde{Z}_t + 1.8803\widetilde{\mu}_t$$

Notice that the solution for $\widetilde{K}_{t+1}$, $\widetilde{Y}_t$, $\widetilde{C}_t$, $\widetilde{I}_t$, $\widetilde{H}_t$, $\widetilde{r}_t$, and $\widetilde{w}_t$, coincides exactly with the solution of the real (non-monetary) model in Hansen (1985). Upon reflection this is not surprising since the previous system of equations can be broken down in two blocks, as follows:

Block 1

$$\widetilde{Y}_t = \widetilde{Z}_t + \theta \widetilde{K}_t + (1 - \theta) \widetilde{H}_t$$

$$\widetilde{w}_t = \widetilde{Y}_t - \widetilde{H}_t$$

$$\widetilde{r}_t = \widetilde{Y}_t - \widetilde{K}_t$$

$$\delta \widetilde{I}_t = \widetilde{K}_{t+1} - (1 - \delta) \widetilde{K}_t$$

$$\overline{Y} \widetilde{Y}_t = \overline{C} \widetilde{C}_t + \overline{I} \widetilde{I}_t$$

$$\widetilde{C}_t = \widetilde{w}_t$$

$$0 = \mathbb{E}_t \left\{ \widetilde{C}_{t+1} - \widetilde{C}_t - \beta \overline{r} \widetilde{r}_{t+1} \right\}$$

$$\widetilde{Z}_{t+1} = \gamma \widetilde{Z}_t + \varepsilon_{t+1} \quad \text{with} \quad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

Block 2

$$\widetilde{M}_t^r = \widetilde{C}_t - \frac{\beta}{\mu} \widetilde{i}_t$$

$$\widetilde{M}_t^r = \widetilde{M}_{t-1}^r + \widetilde{\mu}_t - \widetilde{\pi}_t$$

$$0 = \mathbb{E}_t \left\{ \beta \widetilde{C}_{t+1} + \beta \widetilde{\pi}_{t+1} - \mu \widetilde{C}_t + (\mu - \beta) \widetilde{M}_t^r \right\}$$

$$\widetilde{\mu}_{t+1} = \alpha \widetilde{\mu}_t + \eta_{t+1} \quad \text{with} \quad \eta_{t+1} \sim \mathcal{N}(0, \sigma_\eta^2)$$

Notice that Block 1 is exactly the same as the one in Hansen's model. This block of 8 equations can be solved for $\widetilde{K}_{t+1}$, $\widetilde{Y}_t$, $\widetilde{C}_t$, $\widetilde{I}_t$, $\widetilde{H}_t$, $\widetilde{r}_t$, $\widetilde{w}_t$, and $\widetilde{Z}_t$ (8 variables). Given the solution for $\widetilde{C}_t$ obtained from Block 1 we can solve Block 2 (4 eq.) for $\widetilde{M}_t^r$, $\widetilde{i}_t$, $\widetilde{\pi}_t$, and $\widetilde{\mu}_t$ (4 variables). Remark. Separability of real balances in the utility function is important for this result.

Notice that the only real variable that is affected by money growth is M^r (real balances). When this happens we say that money is superneutral. It is natural to expect that M^r is affected by money growth since we know that money growth affects the nominal interest rate, which is a determinant of the demand for real balances.

Percent Standard Deviations

Real Money	0.9587	(0.088)	0.53
Output	1.8029	(0.232)	1.00
Consumption	0.5182	(0.082)	0.29
Investment	5.7645	(0.731)	3.20
Labor	1.3733	(0.174)	0.74
Rental Rate	1.8373	(0.232)	1.02
Wage	0.5182	(0.082)	0.29
Nominal Interest Rate	0.831	(0.064)	0.46
Gross Inflation	1.6503	(0.119)	0.92

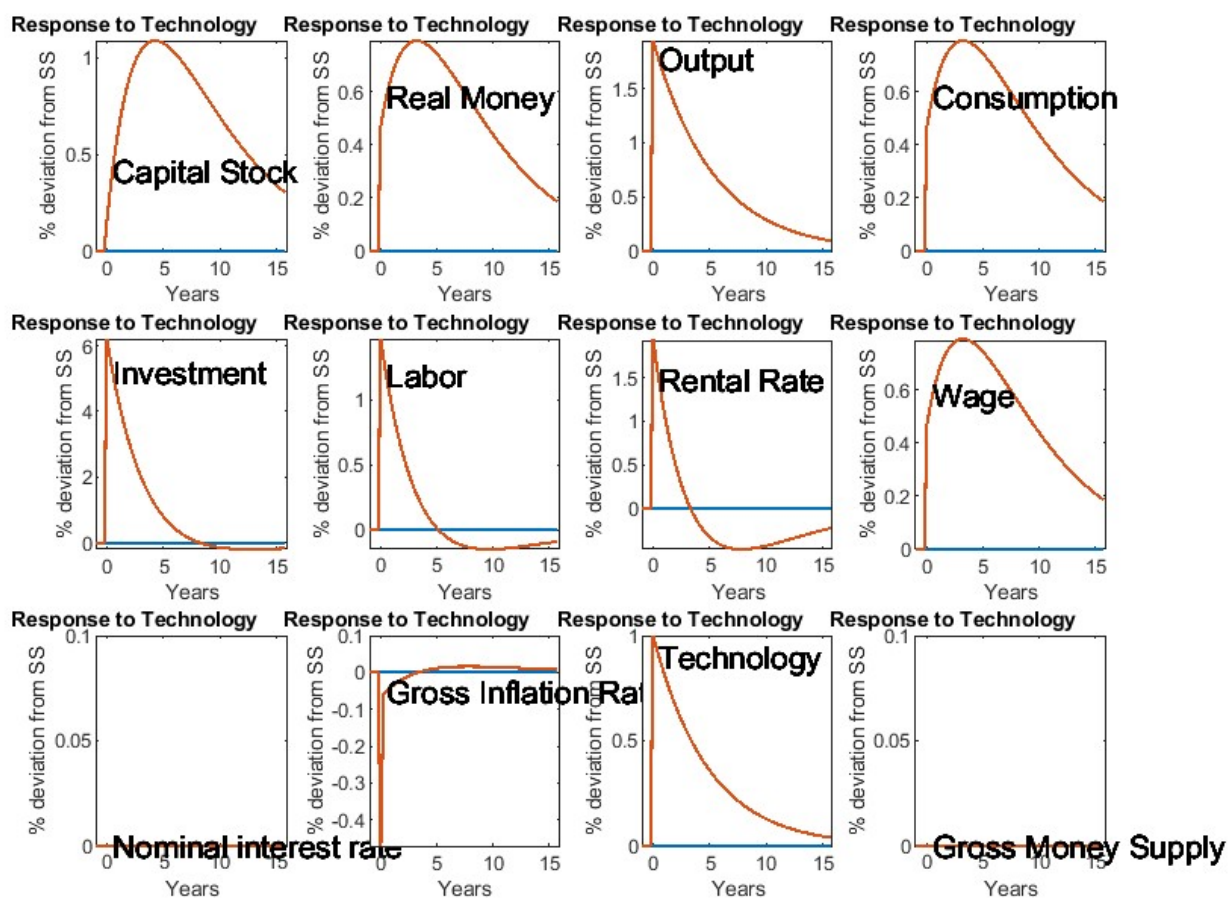
Standard errors in parenthesis; 100 simulations, 115 periods each. Third column: s.d. relative to output.

Contemporaneous Correlations with Output

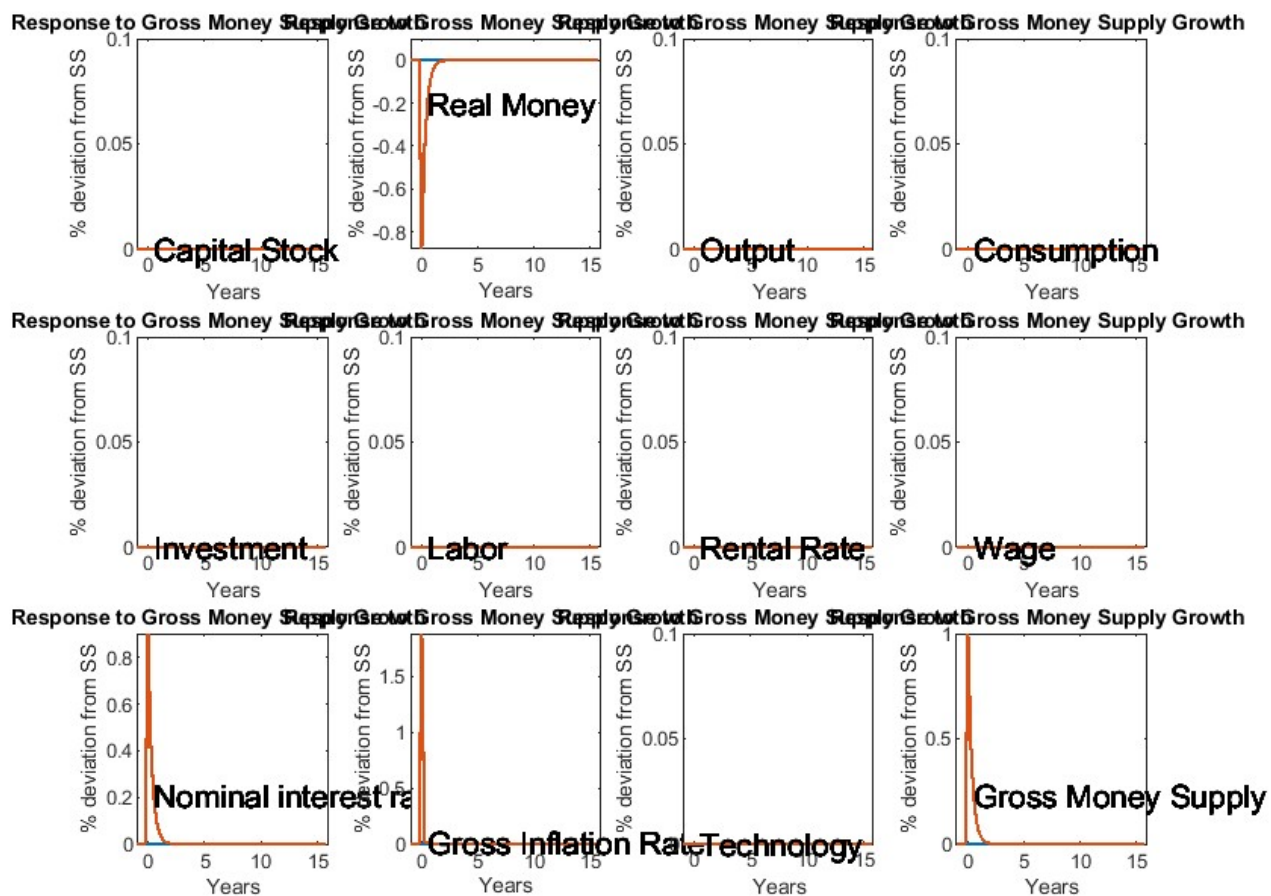
Real Money	0.46	(0.11)
Output	1.00	(0.00)
Consumption	0.88	(0.02)
Investment	0.99	(0.00)
Labor	0.98	(0.00)
Rental Rate	0.96	(0.01)
Wage	0.88	(0.02)
Nominal Interest Rate	0.01	(0.14)
Gross Inflation	−0.1	(0.1)

Standard errors in parenthesis; 100 simulations, 115 periods each.

Impulse Responses: Technology Shock



Impulse Responses: Money Supply Shock



A different version of the model

Government

$$G_t = \frac{M_t - M_{t-1}}{p_t}$$

$$\ln G_{t+1} = (1 - \phi) \ln G + \phi \ln G_t + \xi_{t+1}$$

Remark 1. Interpret G_t as the fiscal deficit at t , and assume all the deficit is monetized.

Remark 2. Drop the lump-sum transfers from the budget constraint of the household.

Remark 3. We assume G_t is exogenous and M_t is endogenous (fiscal dominance)

Now the resource constraint becomes:

$$Y_t = C_t + I_t + G_t$$

In steady state, seigniorage ($S_t \equiv \frac{M_t - M_{t-1}}{p_t}$) is:

$$\begin{aligned}\bar{S} &= \frac{\bar{\mu} - 1}{\bar{\mu}} \bar{M}^r = \frac{\bar{\mu} - 1}{\bar{\mu}} \frac{a\bar{C}}{1 - \frac{\beta}{\bar{\mu}}} \\ &= \frac{\bar{\mu} - 1}{\bar{\mu} - \beta} a\bar{C}\end{aligned}$$

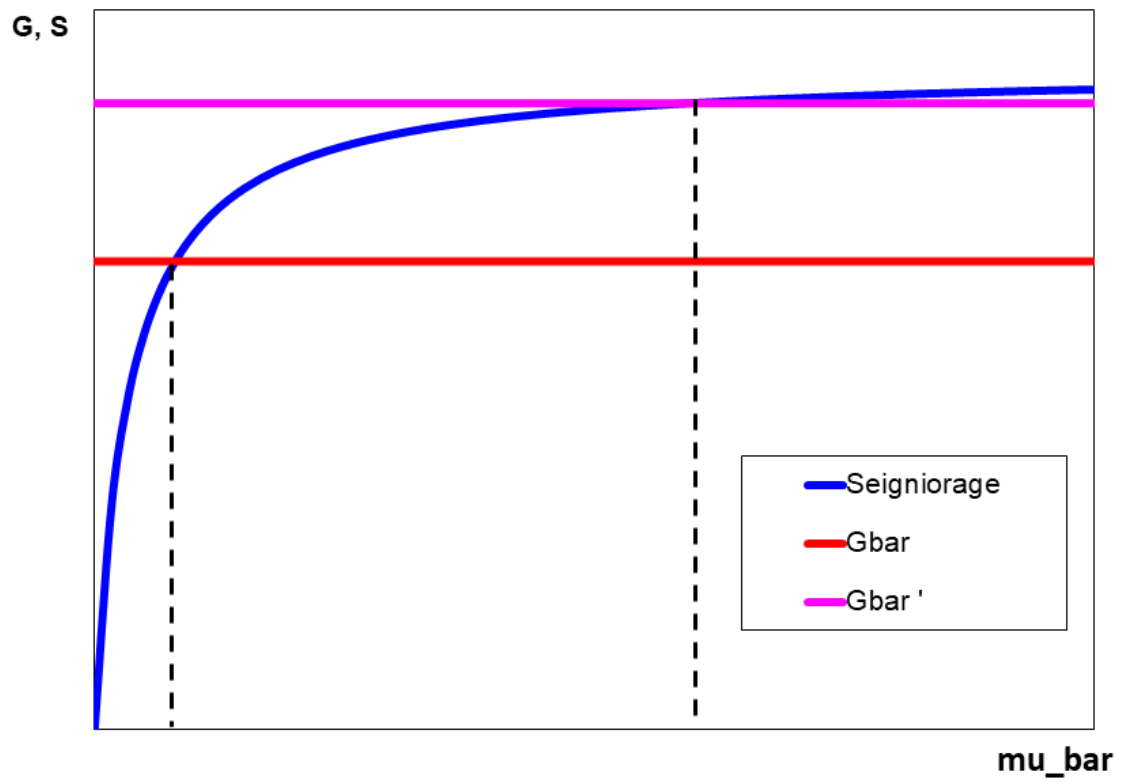
The steady-state value of G_t is:

$$\bar{G} = G$$

The equilibrium value of $\bar{\mu}$ solves: $\bar{G} = \bar{S}$. Then:

$$G = \frac{\bar{\mu} - 1}{\bar{\mu} - \beta} a\bar{C}$$

Remark. In this model, $\bar{C}$ is independent of $\bar{\mu}$ and G .



Appendix
Money in the Utility Function
when the nominal interest rate is zero

From the FOCs w.r.t. c_t and m_t we get:

$$u_c(t) = u_m(t) + \beta \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\} \quad (1)$$

From the FOCs w.r.t. c_t and b_t we get:

$$u_c(t) = \beta (1 + i_t) \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\} \quad (2)$$

Suppose $i_t = 0$. Then, (2) becomes:

$$u_c(t) = \beta \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\} \quad (3)$$

Substituting (1) into (3) we get:

$$u_m(t) + \beta \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\} = \beta \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\}$$

Then:

$$u_m(t) = 0$$

The expression above implies that, when $i_t = 0$, the household would demand real balances until satiation (zero marginal utility).

Notice, however, that the expression above cannot hold if the utility function always displays $u_m > 0$, like in our logarithmic case. What's happening? Well, with $i_t = 0$ and $u_m > 0$, the household could borrow like crazy ($b_t/p_t \rightarrow -\infty$) and use the proceeds to increase real balances ($m_t/p_t \rightarrow +\infty$), and this operation would always increase utility ($u_m > 0$). Hence, the maximization problem of the household would have no solution.

References

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Cooley, T. and Hansen, G. (1989). "The Inflation Tax in a Real Business Cycle Model," *The American Economic Review*, Vol. 79, No. 4 (September), pp. 733-748.